

Description of the NHANES dataset

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Introduction

The National Health and Nutrition Examination Survey (NHANES) is a program conducted by the National Center for Health Statistics (NCHS) to assess the health and nutritional status of the U.S. population (Centers for Disease Control and Prevention 2024). NHANES combines interviews, physical examinations, and laboratory measurements to provide information on a wide range of health indicators.

This report provides a descriptive overview of the NHANES sample across the survey waves included in this analysis. The sample is described according to its size, gender, ethnicity, and average systolic blood pressure (SBP).

NHANES Sample composition by gender and ethnicity

Table 1: Sample size and gender distribution by NHANES wave

NHANES wave	Sample size	Males (n)	Males (%)	Females (n)	Females (%)
2013-2014	10175	5003	49.2	5172	50.8
2015-2016	9971	4892	49.1	5079	50.9
2017-2018	9254	4557	49.2	4697	50.8

The number of participants was almost similar across the three NHANES waves included in the analysis, with the 2013-2014 wave having the most participants. Table 1 summarizes the sample size and gender distribution for each wave. Overall, the gender distribution is similar between waves, with a relatively even split between males (about 49%) and female (about 51%).

Table 2: Ethnicity distribution by NHANES wave

Ethnicity	2013–2014	2015–2016	2017–2018
White	3674 (36.1%)	3066 (30.7%)	3150 (34%)
Black	2267 (22.3%)	2129 (21.4%)	2115 (22.9%)
Mexican	1730 (17%)	1921 (19.3%)	1367 (14.8%)
Asian	1074 (10.6%)	1042 (10.5%)	1168 (12.6%)
Other Hispanic	960 (9.4%)	1308 (13.1%)	820 (8.9%)
Other	470 (4.6%)	505 (5.1%)	634 (6.9%)

The ethnic composition of the sample is presented in Table 2. The distribution of ethnic groups between survey waves was about the same. White participants represented the largest ethnic group across the waves, followed by Black.

Systolic Blood Pressure

Table 3: Summary statistics for average systolic blood pressure by NHANES wave

NHANES Wave	Minimum	Maximum	Median	Mean	Standard deviation
2013-2014	64.7	228.7	115.3	118.3	18.1
2015-2016	74.0	231.3	117.3	120.4	18.4
2017-2018	72.7	234.0	118.0	121.7	20.3

The distribution of average systolic blood pressure across the NHANES waves is summarized in table 3. The mean average systolic blood pressure slightly increased with each wave. The median values followed a similar pattern.

Conclusion

Overall, the NHANES samples were relatively similar in size across the three waves, gender and ethnic composition was also relatively similar. The mean average systolic blood pressure varied somewhat between waves with the 2013-2014 wave having the lowest mean average SBP. These descriptive results provide an overview of the NHANES dataset and the main characteristics of some important variables for the analysis.

References

Centers for Disease Control and Prevention. 2024. *National Health and Nutrition Examination Survey*. National Center for Health Statistics. <https://www.cdc.gov/nchs/nhanes/>.